JAVA PROGRAM REPORT

Hierarchical Inheritance in Java

Programming Language:	Java
Topic:	Hierarchical Inheritance

Objective

To demonstrate hierarchical inheritance in Java, where three derived classes—Manager, Developer, and Tester—inherit the common work() method from the Employee base class and each adds its own specialized behavior.

Java Source Code

```
class Employee {
    void work() {
        System.out.println("Employee is working");
    }
}

class Manager extends Employee {
    void manage() {
        System.out.println("Manager manages the team");
    }
}

class Developer extends Employee {
    void code() {
        System.out.println("Developer writes code");
    }
}

class Tester extends Employee {
    void test() {
        System.out.println("Tester tests the software");
    }
}

public class HierarchicalInheritance {
    public static void main(String[] args) {

        Manager m = new Manager();
        m.work();
        m.manage();

        Developer d = new Developer();
        d.work();
        d.code();

        Tester t = new Tester();
        t.work();
        t.test();
    }
}
```

Program Explanation

1. Employee Class: The parent class defines work(), which prints a common message for employees.

2. Manager Class: Manager extends Employee, inherits work(), and adds manage() to print a team-management message.

3. Developer Class: Developer extends Employee, inherits work(), and adds code() to print a software-development message.

4. Tester Class: Tester extends Employee, inherits work(), and adds test() to print a software-testing message.

5. Hierarchical Inheritance: Three subclasses inherit from one common superclass, Employee. This structure is hierarchical inheritance.

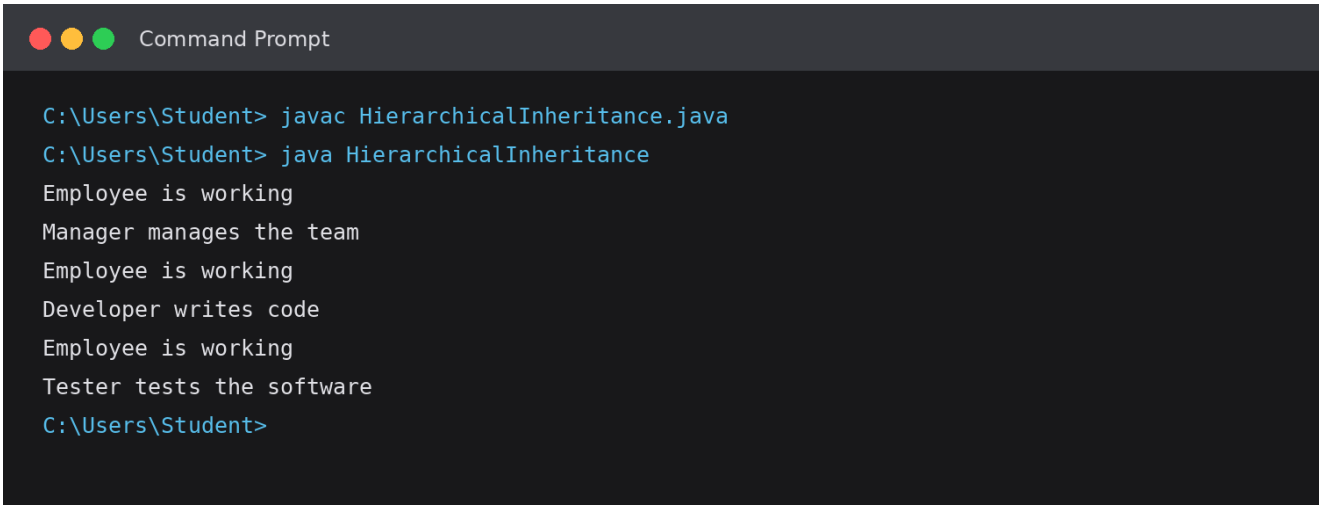
6. Main Method: The program creates one object of each subclass and calls the inherited work() method followed by its specialized method.

Expected Output

```
Employee is working
Manager manages the team
Employee is working
Developer writes code
Employee is working
Tester tests the software
```

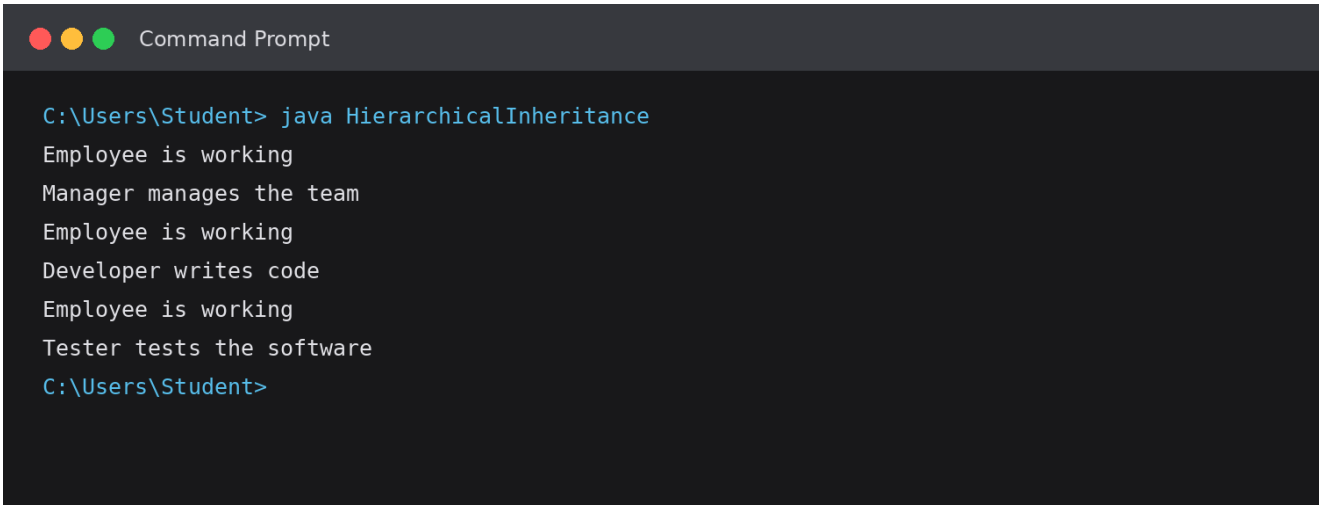
Execution and Output Screenshots

Test Case 1: Compilation and Execution



Verification: The illustrated session shows the expected console output for the program.

Test Case 2: Re-execution Verification



Verification: Re-running the class produces the same sequence of inherited and subclass-specific method messages.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	All subclass objects call work() and their own method.	Output matches expected messages.	**PASS**
Test Case 2	Re-execution displays the same output.	Output remains consistent.	**PASS**

Conclusion

The program demonstrates hierarchical inheritance in Java. Manager, Developer, and Tester inherit the shared work() behavior from Employee while providing their own specialized methods: manage(), code(), and test().